

Surprise Quiz 06

Q1. Signal handlers cannot be provided for which of these signals

- A. SIGKILL
- B. SIGSTOP
- C. SIGINT
- D. SIGCHLD

Q2. Which of the following is used to retrieve the actual numerical exit status code returned by a child process?

- A. WIFEXITED(...)
- B. WEXITSTATUS(...)
- C. wait(...)
- D. waitpid(...)

Q3. How many times the switch from kernel space to user space and user space to kernel space would happen while invoking signal handler provided by a user?

- A. Once for both
- B. Twice for both
- C. Thrice for both
- D. Switch to kernel mode never happens

Q4. Exactly 10 seconds after the program starts executing, what is the status of the child process?

- A. Zombie process
- B. Orphan process
- C. Terminated process without an entry in PCB
- D. Suspended process

```
main() {  
    int ret = fork();  
    if (ret == 0) {  
        sleep(100);  
        exit(0);  
    } else if (ret > 0) {  
        exit(0);  
    }  
    return 0;  
}
```

Q5. Parent process forks a child process which eventually calls exec(). Which of the following is correct after the call to exec in child?

- A. Parent and child only share the .text segment
- B. Parent and child are not sharing any part of their address space
- C. PID of the child process changes to a new value.
- D. There will now be three processes (parent, original child, and new child)

Q6. Exactly 50 seconds after the program starts executing, what is the status of the child process?

- A. Zombie process
- B. Orphan process
- C. Terminated process without an entry in PCB
- D. Suspended process

```
main() {  
    int ret = fork();  
    if (ret == 0) {  
        exit(0);  
    } else if (ret > 0) {  
        sleep(100);  
    }  
    return 0;  
}
```

Answer Key:

Q1: A and B

Q2: B

Q3: B

Q4: B

Q5: B

Q6: A